

A Retrieval-Augmented AI Assistant for Erasmus+ Exchange Support at UL FRI

Sara Ivanovska, Aleksa Sibinović and Hristijan Milanovski

Abstract

We present a retrieval-augmented AI assistant for Erasmus+ exchange support at UL FRI, retrieving information from curated official sources including university materials, Erasmus+ guidance documents, and resources obtained directly through the UL FRI Erasmus coordination process. Documents are chunked, embedded using a sentence-transformer model, indexed with FAISS, and retrieved using Maximum Marginal Relevance search before answer generation with local open-weight language models. The system is evaluated on a manually curated benchmark of 210 Erasmus-related questions using three complementary approaches: BERTScore semantic evaluation, LLM-as-judge assessment via the Gemini API, and manual human evaluation, comparing two local open-weight models, Llama 3.1-8B and Gemma 3-4B, under initial ($k = 3$) and tuned ($k = 7$) retrieval configurations. Results show that retrieval grounding reduces hallucinations by nearly an order of magnitude compared to a Gemini no-retrieval baseline, with manual evaluation scores improving from 1.88 to 2.32 for the best RAG configuration, and that retrieval depth matters more than model scale in this domain.

Keywords

Retrieval-Augmented Generation, Domain-Specific Assistant, Large Language Models, Question Answering, Student Office

Advisors: Aleš Žagar

Introduction

Students preparing for Erasmus exchange at UL FRI must navigate a large amount of administrative information spread across faculty pages, University of Ljubljana documents, official Erasmus+ guidance, PDF forms, and partner university catalogues. Questions are often procedural and context-dependent, covering deadlines, Learning Agreements, ECTS recognition, financial support, visa requirements, and post-mobility documentation, and rarely have a single authoritative source. General-purpose language models can produce fluent and plausible sounding answers, but without access to institution-specific sources they often hallucinate deadlines, funding amounts, or eligibility conditions. In an administrative context, such errors can directly harm a student's exchange plans.

We address this by developing a domain-specific AI assistant grounded in curated official and institution-specific sources. Our approach is based on retrieval-augmented genera-

tion (RAG): the system retrieves relevant document chunks from a curated Erasmus knowledge base and uses a large language model to generate an answer grounded in the retrieved content. This combines the flexibility of conversational question answering with the factual grounding of official documents, reducing the risk of unsupported or hallucinated responses.

We evaluate whether retrieval grounding improves factual reliability compared to a Gemini 2.5 Pro no-retrieval baseline, and compare two local open-weight models, Llama 3.1-8B and Gemma 3-4B, under both an initial retrieval configuration ($k=3$) and a tuned one ($k=7$).

Dataset

Since no pre-existing dataset exists for this domain, we constructed our own knowledge base from publicly available institutional and Erasmus-related sources. The dataset combines publicly available university materials with Erasmus-specific

resources obtained through direct interaction with the UL FRI Erasmus coordination process. Institutional sources include the UL FRI website and the University of Ljubljana central website. We additionally incorporated official Erasmus+ guidance documents covering financial support, Online Learning Agreements (OLA), student rights and obligations, visa and residence procedures, doctoral mobility, traineeships, and UL FRI partner institutions. Structured course-guidance files for selected partner universities were also prepared to support questions related to course selection and exchange planning.

A particularly valuable part of the dataset was obtained through direct engagement with the Erasmus coordination process at UL FRI. We attended the official Erasmus exchange kickoff meeting for the 2026/27 academic year, where we interacted with the main coordinator, Vesna Gračner, and obtained the official presentation used during the session. We also systematically recorded and transcribed student questions raised during the Q&A session, together with the coordinator’s answers. These question-answer pairs capture realistic information needs and practical procedural clarifications that are not consistently documented in official written materials. The collected materials were organised by topic, covering admissions, study programmes, visa and residence procedures, student life, and Erasmus exchange. Additionally the Q&A content was used both to enrich the knowledge base and to construct the evaluation benchmark described in the Results section.

Methods

The assistant is built around a retrieval-augmented generation (RAG) pipeline consisting of two stages: an offline indexing stage that preprocesses and stores the knowledge base, and an online inference stage that retrieves relevant passages and generates a grounded answer. The overall architecture is illustrated in Figure 1.

Document indexing

All knowledge-base documents are loaded and parsed before any queries are processed. Plain text files are read directly, while PDF documents are parsed page by page.

Documents are split into overlapping chunks using a recursive character splitter. We experimented with chunk sizes of 300, 500, and 700 characters and found that 500 characters with a 50-character overlap offered the best balance between retrieval precision and contextual completeness. Smaller chunks frequently fragmented multi-step procedural answers, while larger chunks introduced irrelevant context. Each chunk is embedded into a dense vector representation using the *all-MiniLM-L6-v2* sentence-transformer model [1], with output vectors L2-normalised to unit length. The resulting embeddings are stored in a FAISS flat index [2], saved to disk and reused at query time.

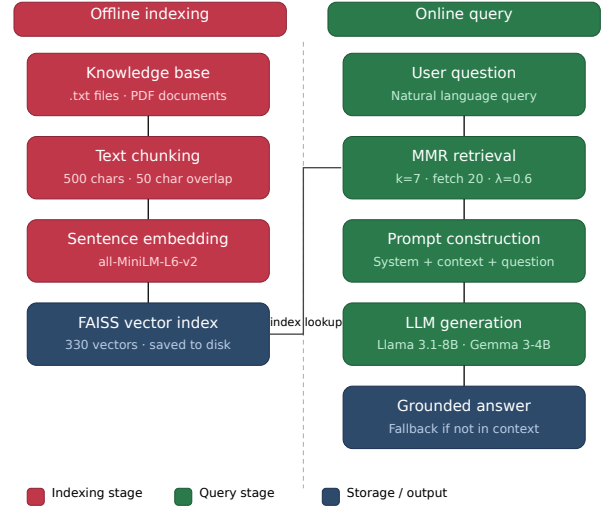


Figure 1. Overview of the RAG pipeline. The offline phase builds a FAISS vector index from the curated knowledge base; the online phase retrieves relevant chunks and generates a grounded answer.

Retrieval and generation

At query time, the FAISS index is used as the backbone of a retriever configured with Maximum Marginal Relevance (MMR) search. MMR selects chunks that are both relevant to the query and mutually diverse, reducing redundancy when multiple passages cover the same topic. We tuned the retrieval configuration by varying the number of retrieved chunks k (3, 5, 7), chunk size (300, 500, 700 characters), search type (MMR vs. similarity), and embedding model (MiniLM vs. mpnet), evaluating each combination on 50 questions. As shown in Table 1, $k=7$ with MMR consistently produced the highest F1 and key-information coverage with the lowest fallback rate, and was adopted as the final configuration. In the MMR setup, 20 candidate chunks are initially fetched (fetch_k=20), the top 7 are selected by balancing relevance and diversity ($\lambda=0.6$, where higher values favour relevance over diversity). Both the baseline ($k=3$) and tuned ($k=7$) configurations were evaluated on the full 210-question benchmark to quantify the impact of retrieval depth.

Configuration	Avg F1	Key info	Fallback
$k=3$, chunk=500, MMR	43.8%	90.0%	8.0%
$k=5$, chunk=500, MMR	47.8%	92.0%	4.0%
$k=7$, chunk=500, MMR	51.6%	92.0%	2.0%
$k=3$, chunk=300, MMR	39.3%	78.0%	10.0%
$k=3$, chunk=700, MMR	40.9%	78.0%	10.0%
$k=5$, similarity	45.0%	86.0%	4.0%
$k=5$, MMR, mpnet	47.6%	88.0%	4.0%

Table 1. Retrieval configuration comparison on 50 questions. Bold indicates the selected final configuration.

The retrieved chunks are concatenated into a context string, each prefixed with its source path, and injected into a structured prompt that instructs the model to answer using only the

provided passages. If the context does not contain a sufficient answer, the model responds with a fixed fallback message directing the student to the UL FRI website or student office, rather than generating an unsupported response.

Answer generation is handled locally on the ARNES HPC cluster using two open-weight models: *Llama 3.1-8B-Instruct* and *Gemma 3-4B-IT*. Llama 3.1-8B served as the primary model and consistently outperformed Gemma 3-4B on questions requiring exact procedural or administrative details specific to UL FRI. The full pipeline is assembled as a single LangChain LCEL chain: the retriever feeds formatted context into the prompt template, which is passed to the language model and parsed into plain text answer.

Results

We evaluated the system on a manually curated benchmark of 210 Erasmus-related questions covering procedural, administrative, legal, and general informational topics. Questions were constructed to span varying levels of retrieval dependence, from those requiring exact institutional information to broader questions that can be partially answered from general world knowledge. The benchmark also includes deliberately vague and ambiguous cases to reflect realistic student interactions.

Two models were evaluated within the same RAG pipeline: Llama 3.1-8B and Gemma 3-4B, each under an initial retrieval configuration ($k = 3$) and the tuned configuration ($k = 7$).

BERTScore evaluation

We tested 14 different BERT score metrics, of the families DeBERTa v1 [3], DeBERTa v2 [4], DeBERTa v3 [5], XLNet [6], ELECTRA [7] and RoBERTa [8] using bootstrap settings: 10,000 resamples, 95% percentile confidence intervals, seed 42. The highest absolute-F1 scorer across all model configurations was roberta-large with mean F1 0.8787. On the other hand, the lowest absolute-F1 scorer was xlnet-large-cased with mean F1 0.5541. We report results for the best-performing scorer, roberta-large, as shown in Table 2. Full results for all 14 configurations are available in the project repository.

Model	Not tuned	Tuned	Δ
Gemini 2.5 Pro baseline	0.8401	—	—
Gemma	0.8853	0.8929	0.0076
Llama	0.8861	0.8892	0.0031

Table 2. Mean F1 scores for tuned and non-tuned models evaluated with the roberta-large scorer.

Moreover, Gemma bootstrap CIs exclude zero for 14/14 scorer models, while Llama bootstrap CIs exclude zero for 5/14 scorer models. These results suggest that the observed effect is robust and consistently statistically significant for Gemma across all scorer models, whereas evidence for the effect in Llama is much weaker and less consistent.

LLM-as-judge evaluation

We used the Gemini API to evaluate how well the five models answered each of the 210 questions. The results in Table 3 indicate a noticeable bias toward the Gemini baseline responses. This is likely because the Gemini baseline system prompt explicitly included the university websites uni-lj.si and fri.uni-lj.si, providing the model with direct access to some available information. The Gemini-generated answers frequently referenced links and documents that appeared plausible and contextually appropriate at first glance. As a result, the evaluation process often failed to distinguish between superficially convincing answers and genuinely correct ones, which likely contributed to inflated evaluation scores compared to the other models.

The Gemini API evaluation system assigns scores in four bands: Correct, Mostly correct, Partially correct, and Incorrect.

Model	Corr.	Mostly. Corr.	Partiall. Corr.	Incorrect
Gemini 2.5 Pro baseline	55.2%	17.1%	15.7%	11.9%
Gemma 3 RAG, $k = 3$	29.5%	31.0%	33.8%	5.7%
Gemma 3 RAG, $k = 7$	32.9%	37.6%	25.7%	3.8%
Llama 3.1 RAG, $k = 3$	32.5%	38.6%	25.4%	3.5%
Llama 3.1 RAG, $k = 7$	39.5%	32.4%	22.4%	5.7%

Table 3. Performance comparison across models in percentages, indicating a clear bias towards the Gemini baseline.

Human evaluation

While automated metrics capture lexical and semantic similarity, they cannot reliably detect subtle hallucinations or assess whether an answer is practically useful for a student. We therefore conducted a manual evaluation using a four-level scoring scheme, in which **3** indicates a correct and complete answer, **2** indicates a mostly correct answer but missing important details, **1** indicates a partially correct or vague answer, and **0** indicates an incorrect or unusable answer.

During manual evaluation, responses were additionally annotated for hallucinations. A hallucination was defined as any unsupported or fabricated information generated by the model, either as part of an otherwise correct answer or as the entire response. Fallback responses indicating that sufficient information was unavailable and directing students to the relevant office were not considered hallucinations and could still receive positive scores if the fallback was appropriate for the question.

The Gemini baseline was first evaluated automatically using an LLM-as-a-judge approach, but the scores appeared overly optimistic and did not sufficiently penalize unsupported procedural claims or subtle hallucinations. We therefore manually re-evaluated the baseline using the same protocol applied to all RAG configurations. This reduced the baseline average score from 2.09 to 1.88 and decreased the proportion of fully correct answers from 55.7% to 45.2%. Notably, many hallucinations were not detected automatically because the model

failed to recognize outdated or institution-specific information.

Model	Avg. score	Score ≥ 2	Score 3	Hallucinations
Gemini 2.5 Pro baseline	1.88	61.4%	45.2%	67 (31.9%)
Llama 3.1 RAG, $k = 3$	2.13	70.0%	60.5%	9 (4.3%)
Gemma 3 RAG, $k = 3$	1.93	65.2%	53.3%	13 (6.2%)
Llama 3.1 RAG, $k = 7$	2.32	78.1%	65.7%	11 (5.2%)
Gemma 3 RAG, $k = 7$	2.27	76.7%	63.8%	11 (5.2%)

Table 4. Manual evaluation results across the baseline and RAG configurations.

Table 4 indicates that retrieval primarily improved factual grounding and answer reliability rather than only raw correctness scores. Although the baseline often produced fluent and detailed answers for broader informational questions, it was frequently overconfident when handling vague or institution-specific procedures. This was especially visible when information differed across multiple partially outdated sources, such as Erasmus deadlines, travel support amounts, or administrative requirements. In several cases, the model hallucinated non-existent links or mixed outdated and current procedural information within the same answer.

Importantly, many baseline answers that appeared mostly correct still included additional unsupported details that could confuse students or contradict official procedures. In contrast, retrieval grounding reduced hallucination rates by nearly an order of magnitude across all evaluated configurations. The results also suggest that retrieval depth had a stronger impact than the choice of local language model. Increasing the number of retrieved chunks from $k = 3$ to $k = 7$ consistently improved both Llama and Gemma, indicating that broader contextual coverage helped the models recover important procedural details and constraints that were often omitted in the smaller retrieval setup.

Interestingly, the tuned Gemma configuration approached the performance of the larger Llama model despite having fewer parameters, suggesting that retrieval quality and context selection may play a more important role in this domain than model scale alone.

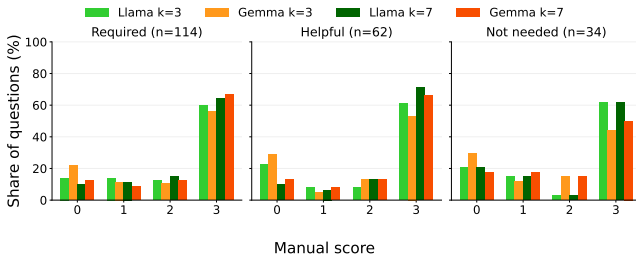


Figure 2. Distribution of manual evaluation scores (0–3) across retrieval dependency categories and RAG configurations. Questions were grouped by how strongly they depended on retrieval: *required*, *helpful*, and *not needed*. The figure shows the percentage of questions assigned each manual score (0–3) for both models and retrieval settings.

Figure 2 shows that retrieval-dependent questions benefit

the most from increasing retrieval depth. Both models produce more fully correct answers (score 3) and fewer failures (score 0) in the *required* category after increasing retrieval from $k = 3$ to $k = 7$. This suggests that additional retrieved context primarily improves access to institution-specific procedural details that are difficult to answer from parametric model knowledge alone.

The effect is particularly pronounced for Gemma 3-4B, whose performance gap relative to Llama 3.1-8B becomes noticeably smaller after tuning. This indicates that smaller models are more sensitive to retrieval quality and context coverage.

In contrast, questions categorized as *not needed* show smaller differences between retrieval configurations, suggesting that broader informational questions depend less on retrieval grounding and more on the underlying language model itself.

Discussion

Base models often produced plausible high-level answers, while retrieval grounding helped recover institution-specific details necessary for practical student guidance. Automatic metrics alone were insufficient to capture this difference, as they rewarded semantically similar but procedurally unreliable answers, making manual evaluation essential for assessing factual grounding and hallucinations accurately.

Most observed errors were not completely incorrect answers, but plausible yet incomplete responses missing deadlines, eligibility constraints, or procedural details specific to UL FRI. In retrieval-grounded configurations, errors were typically caused by ambiguous questions, context-dependent scenarios where the correct answer depends on the student’s specific situation, or cases not explicitly covered in the knowledge base, often resulting in appropriate fallback responses.

The results suggest that smaller local models can achieve competitive performance when paired with sufficiently strong retrieval. Despite the substantial difference in model size, the tuned Gemma configuration approached the performance of the larger Llama model after retrieval depth was increased, indicating that careful retrieval design may provide larger practical gains than model scale alone in closed institutional domains. Future work could extend the knowledge base with automatically scraped and updated sources, introduce a mechanism for detecting and flagging outdated information such as expired deadlines or changed funding amounts, and explore re-ranking or query expansion to further improve retrieval precision.

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